

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of BSc (BC) Examination April-May 2018

Course code & Name: BC 401: Biomolecules and nutrition

Date: 07/05/2018 Day: Monday Time: 10:00 am To 10:20 am Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 10

1. Which of following bonds is/are present in protein secondary structure?
(i) Peptide bond only
(ii) Peptide bond, hydrogen bond and di-sulfide bond
(iii) Peptide bond and hydrogen bond
(iv) Hydrogen bond and di-sulfide bond
2. Vitamin B₂ is also called as _____.
(i) Niacin (ii) Riboflavin
(iii) Thiamine (iv) Rationol
3. Which of the following is/are example non-reducing sugar?
(i) Glucose (ii) Fructose
(iii) Sucrose (iv) (i) and (ii) both
4. Which macromolecule has the highest caloric value?
(i) carbohydrate (ii) lipid
(iii) protein (iv) nucleic acid

5. Which of the following is an example of derived lipids?
- (i) steroids (ii) Terpenes
(iii) Carotenoids (iv) All of these
6. The synthesis of glucose from fat is called _____.
(i) glycolysis (ii) saponification
(iii) Kreb's cycle (iv) Gluconeogenesis
7. Methionine-folate cycle has a role in _____.
(i) DNA synthesis and repair (ii) protein modification
(iii) lipid degradation (iv) lipid biosynthesis
8. Bomb calorimeter is used to determine the calorific value of solids and liquid fuels.
☐ True ☐ False
9. Thromboxane is a member of the family of lipids known as eicosanoids.
☐ True ☐ False
10. Methionine and cysteine both are involved in disulfide bond formation in protein.
☐ True ☐ False

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IV Semester of BSc (BC) Examination April-May 2018

Course code & Name: BC 401: Biomolecules and nutrition

Date: 07/05/2018 Day: Monday Time: 10:21 am To 12:00 pm Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

Q - II Answer the following questions as directed (any two) 04

- (a) Explain in brief about polysaccharide with two example
- (b) Define: phospholipids and sphingolipids
- (c) Write a note on “fat soluble Vitamins”

Q - III Answer the following questions as directed (any two) 06

- (a) Write the differences between saturated and unsaturated fatty acid
- (b) Discuss in brief about physico-chemical properties of amino acids.
- (c) Write a note on “Balance diet”

SECTION - II

Q-IV Answer the following questions as directed (any three) 15

- (a) Classify the proteins based on structural organization and explain the biological functions of proteins.
- (b) Describe in detail about nutritional importance of carbohydrates, fats and proteins.
- (c) What is calorific value? Explain the disease associated with nutrition imbalance.
- (d) What is a steroid? Discuss in detail about different lipid molecules that are used as signaling molecules.